Random Walks

ADVANCED TOPICS IN NETWORK SCIENCE · MODULE 07

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#1 title

Where is the walker after 100 steps?

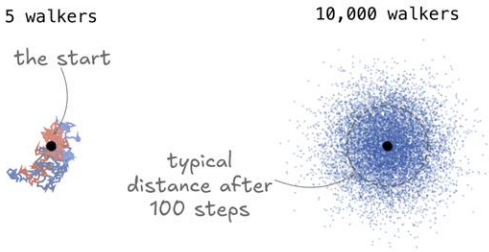
A man walks 1 yard, turns in a random direction, and walks 1 yard again. He repeats this 100 times.

Pearson asked this in *Nature* in 1905.



Far from the start, or near it? Tell your neighbour.

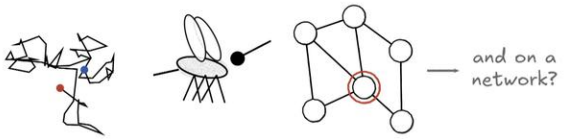
#2 hook (story): Pearson's question, 1905



Rayleigh replied the next week: he had solved it in 1880, for waves

#3 answer: Pearson's walker (the density peaks at the s

Random walks appear in many fields



pollen in water, 1905 | mosquitoes and malaria | the web, 1998

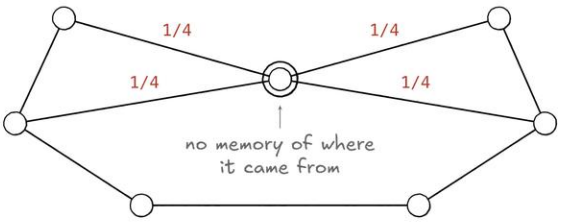
#4 hook: the same walk in three fields

How does it tell us about a network?

But it is a "random" walk, right?

#5 spine 1: the question

What is a random walk?



#6 mechanism: the rule

Pen & Paper

10 minutes · stop at the line on the sheet

#7 Pen & Paper: first block

Random walk simulator

skojaku.github.io/
adv-net-sci/assets/vis/
random-walks/index.html

#8 the simulator (QR code and link)

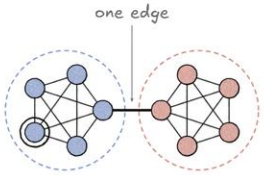
Two communities joined by one edge

How many steps, on average, until the walker crosses to the other community?

A: about 3

B: about 20

C: about 200

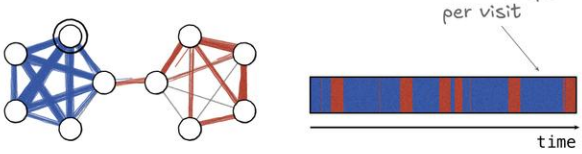


Take a guess.

#9 question to the room (Q2)

OPTIONAL

A short walk stays in its community



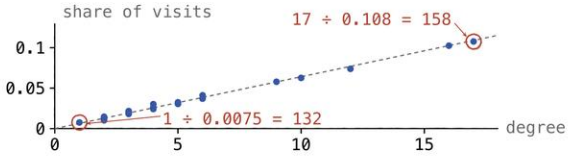
#10 answer to Q2

Short walks show communities. Long walks show degree. Why?

#11 spine 2: the mystery

Degree ÷ share is constant

Karate club: 34 nodes, 78 edges, so 2m = 156



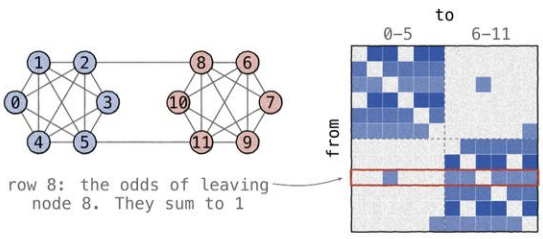
Share of time = degree ÷ 2m

#12 debrief Part 1 (degree ÷ share)

Why does a long walk not depend on the start?

#13 spine 3: part question

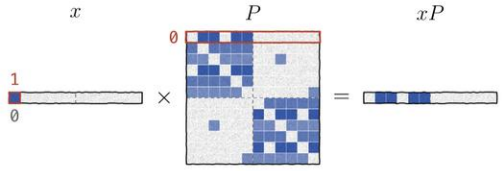
The walk is defined by one matrix



#14 P (the two-community network, the motif of the rest

One step is a multiplication by P

$$x(t+1) = x(t)\,P$$

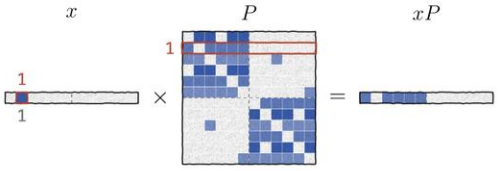


$x = (1, 0, 0, \dots)$: xP is the row of node 0

#15 x P, how to read it (1/3)

One step is a multiplication by P

$$x(t+1) = x(t)\,P$$

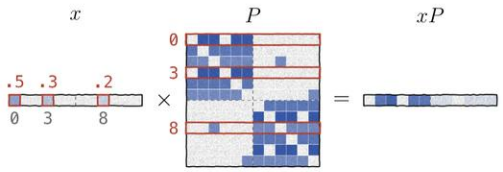


$x = (0, 1, 0, \dots)$: xP is the row of node 1

#16 x P, how to read it (2/3)

One step is a multiplication by P

$$x(t+1) = x(t)\,P$$



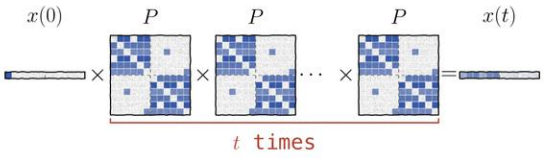
$x = (0.5, 0.05, \ldots)$: xP is a mix of the rows

17

#17 x P, how to read it (3/3)

After t steps

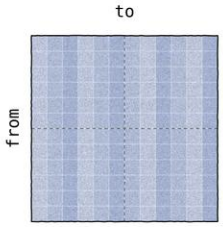
$$x(t) = x(0)\,P^t$$



18

#18 x(0) P^t

Rows of P^t after t steps



$t = 40$

every row is the same

19

#19 P^t as a heat map (slider)

Rows of P^t at $t = 40$

	0	1	2	3	4	5	6	7	8	9	10	11
start 0	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10
start 1	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10
start 2	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10
start 3	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10
start 4	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10
start 5	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10
start 6	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10
start 7	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10
start 8	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10
start 9	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10
start 10	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10
start 11	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10

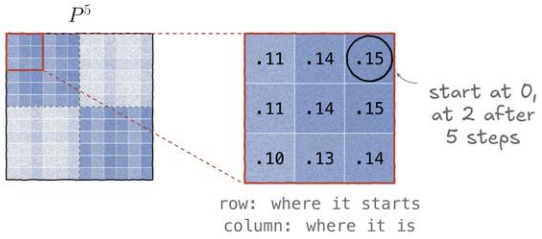
π

Every row becomes the same: π

20

#20 P^t, the numbers at t = 40

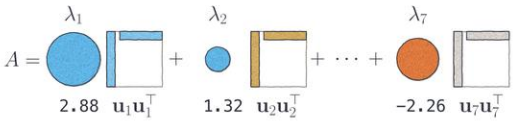
What each entry of P^t says



21

#21 zoom into P^t

Recall: eigendecomposition

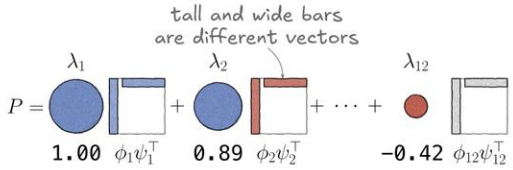


22

#22 recap: eigendecomposition (M06, light touch)

P has an eigendecomposition too

$$P = \sum_k \lambda_k\,\phi_k\,\psi_k^\top$$

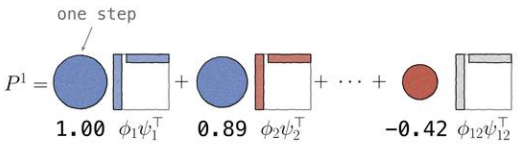


P^t changes only the eigenvalues

23

#23 P as a sum of circles and bars

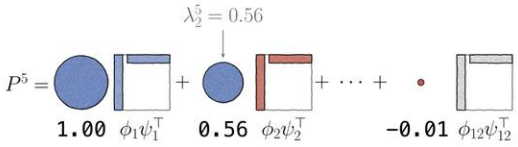
The eigenvalues shrink as t grows



24

#24 P^t as circles (1/3)

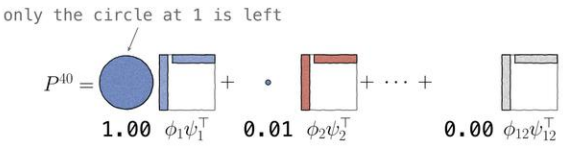
The eigenvalues shrink as t grows



25

#25 P^t as circles (2/3)

The eigenvalues shrink as t grows



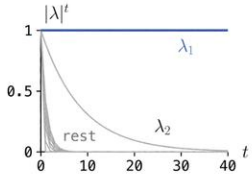
26

#26 P^t as circles (3/3)

Which term stays the longest?

The 12 eigenvalues of P.

After many steps, which term is still there?



Take 20 seconds.

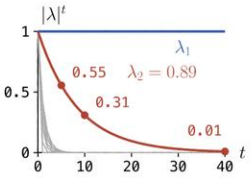
27

#27 Q3 (which term stays the longest)

The second term stays the longest

$\lambda_2 = 0.89$

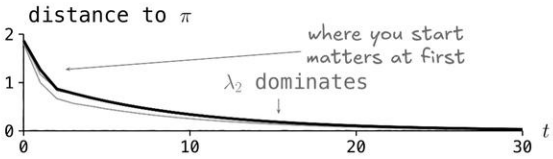
0.89^t: 0.56 at $t = 5$, 0.31 at $t = 10$, 0.01 at $t = 40$



28

#28 A3

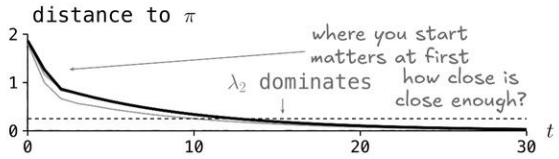
Mixing time



29

#29 mixing time (1/3)

Mixing time

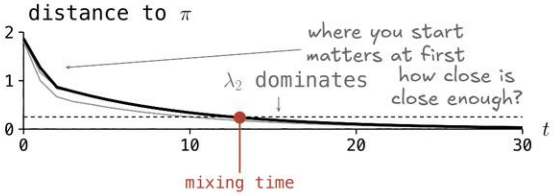


30

#30 mixing time (2/3)

Mixing time

$t_{\text{mix}} = \min\{ t : \max_{x(0)} \|x(t) - \pi\|_1 \le \epsilon \}$



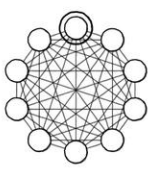
31

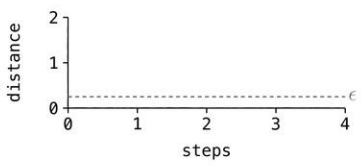
#31 mixing time (3/3)

OPTIONAL

Q: mixing time of a 10-clique?

10 nodes, every pair connected (45 edges)





Take a guess: 1 step, 5 steps, 50 steps.

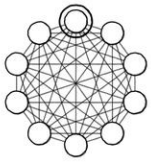
32

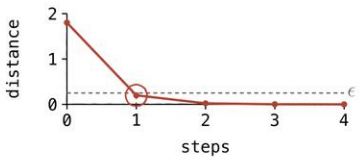
#32 Q4 (clique)

OPTIONAL

A: about 1 step

10 nodes, every pair connected (45 edges)





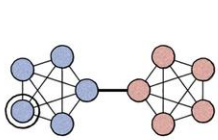
$\lambda_2 = -1/9$: **mixing takes about 1 step**

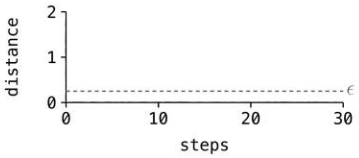
33

#33 A4

Q: mixing time with one bridge?

Two 5-cliques joined by one edge (21 edges)





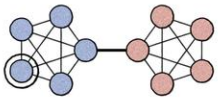
The same, 5 times longer, or 20 times longer?

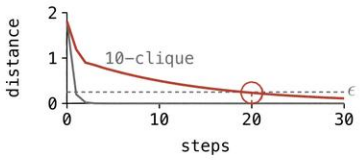
34

#34 Q5 (two cliques)

A: about 20 times longer

Two 5-cliques joined by one edge (21 edges)





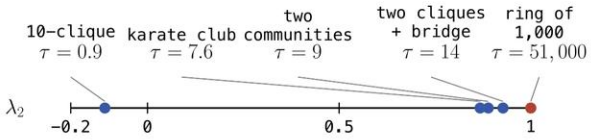
$\lambda_2 = 0.93$: **mixing takes about 20 steps**

35

#35 A5

The mixing time follows from λ_2

$$\tau = \frac{1}{1-\lambda_2}$$



Small gap, long mixing time

36

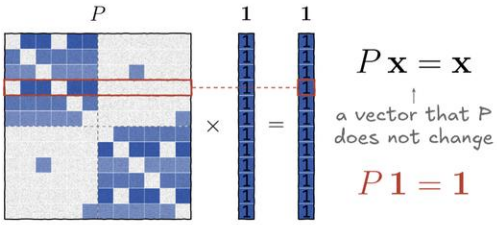
#36 spectral gap (gauge)

One eigenvalue is always 1. Others diminish.

37

#37 spine 4: the fact

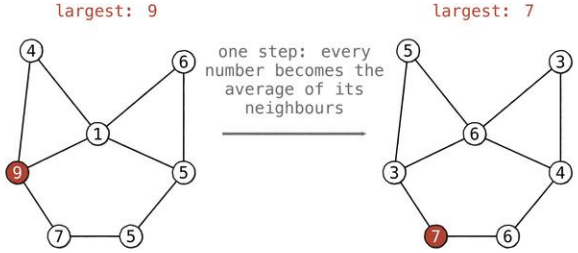
Why 1? Every row of P sums to 1



38

#38 lambda = 1 exists (P 1 = 1)

No eigenvalue is larger than 1

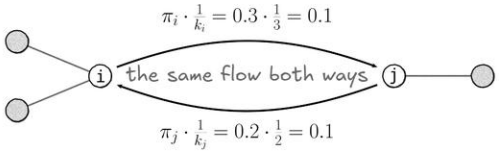


39

#39 no eigenvalue above 1

Detailed balance condition

$$\pi_i P_{ij} = \pi_j P_{ji}$$



$\pi_i = k_i/2m$: **proportional to degree**

40

#40 the eigenvalue 1 gives pi (ties to the simulator's

Pen & Paper

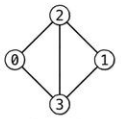
20 minutes

41

#41 Pen & Paper: second block

Tasks

2. Four nodes: write $x(t)$ with the eigenvalues, check $t = 2$
3. Twelve nodes: run one cell, plot the second eigenvector



Part 2

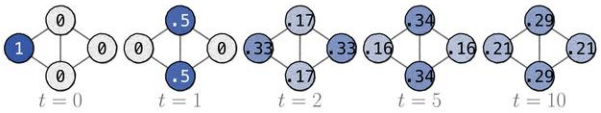
QR code
Part 3
notebook

42

#42 Pen & Paper: instructions

$x(t) = \pi$ + decaying terms

Four nodes; eigenvalues $1,\ 0,\ -\frac{1}{3},\ -\frac{2}{3}$



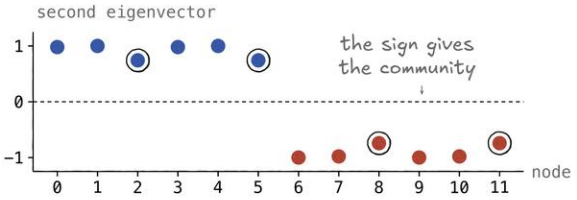
$$x(t) = \pi - 0.5 \cdot 0^t\,\psi_2 + 0.3\,(-\tfrac{2}{3})^t\,\psi_4$$

$t = 2:\quad x(2) = (\frac{1}{3}, \frac{1}{3}, \frac{1}{6}, \frac{1}{6})$

43

#43 debrief Part 2

What is the decaying term?

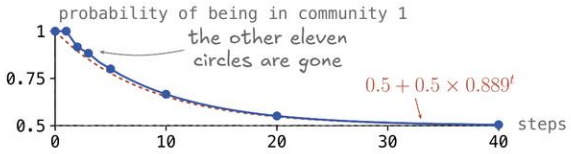


44

#44 debrief Part 3 (second eigenvector)

Early on, the community signal dominates

Walker starts at node 0

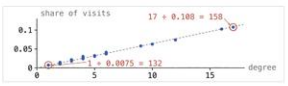


The community signal decays like λ_2^t. Degree remains.

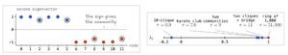
45

#45 debrief Part 3 (what dominates early)

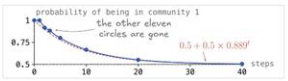
1 The eigenvalue 1 gives the degree



2 The second eigenvalue gives the communities and the mixing time



3 Short walks see communities. Long walks see degree.



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#46 summary (claims with thumbnails)